

# 1 Q3–Q4: Graph Creation and Graph Essentials

## Q3: Graph Creation (10 pts)

The edge list `graph.txt` (attached to this assignment as `graph-1.txt`) contains 26,377 tab-separated lines, each giving one undirected edge. It was loaded into a simple undirected graph (`networkx.Graph`), adding one node per distinct identifier appearing in the file and one edge per line; no isolated nodes were added and the file was not symmetrised a second time. The file contains no self-loops and no duplicate edge in either orientation, so the 26,377 lines yield exactly 26,377 distinct edges.

| Quantity                       | Value  |
|--------------------------------|--------|
| Number of nodes                | 26,728 |
| Number of edges                | 26,377 |
| Number of connected components | 4,104  |

Table 1: Q3: basic counts for the graph built from the edge list.

**Answer.** The graph has **26,728 nodes**, **26,377 edges** and **4,104 connected components**.

The graph is therefore very sparse (mean degree  $2m/n \approx 1.97$ ) and highly fragmented. It is not a forest: a forest on  $n$  nodes with  $c$  components would have  $n - c = 26,728 - 4,104 = 22,624$  edges, whereas the graph has 26,377, so it carries  $m - n + c = 3,753$  independent cycles.

## Q4(a): Largest connected component (10 pts)

Connected components were extracted with `nx.connected_components` and sorted by size in decreasing order; the largest is the first element. Its edge count is that of the *induced* subgraph on those nodes, not the edge count of the whole graph.

| Component      | Nodes  | Edges  |
|----------------|--------|--------|
| Largest (LCC)  | 13,903 | 16,695 |
| Second largest | 69     | 80     |
| Whole graph    | 26,728 | 26,377 |

Table 2: Q4: size of the two largest connected components, against the whole graph.

**Answer.** The largest connected component has **13,903 nodes** and **16,695 edges**. It is a giant component in the usual sense: it holds 52.0% of all nodes and 63.3% of all edges, while the remaining 4,103 components share the other half of the nodes.

## Q4(b): Visualization of the second largest connected component (10 pts)

**Which component, and how it was selected.** The components were sorted by node count in decreasing order and the element at index 1 was taken, i.e. `sorted(nx.connected_components(G), key=len, reverse=True)[1]`. The ten largest component sizes are

13,903, **69**, 30, 30, 29, 26, 23, 21, 21, 20.

**There is no tie for second place:** exactly one component has 69 nodes, so the choice is unambiguous and no tie-breaking rule was needed. (The tie that does exist is at *third* place, where two

distinct components both have 30 nodes; it does not affect this question.) The selected component has **69 nodes** and **80 edges**.

Figure 1 shows it. The layout is Kamada–Kawai, which is deterministic for a given graph (circular initialisation followed by BFGS energy minimisation), so the drawing is reproducible. Node area encodes degree and node colour repeats it via the colour bar. **Node identifiers are deliberately not drawn:** with 69 five-digit identifiers the labels overlap each other and are occluded by the markers in the dense core of the component, which makes the picture less readable rather than more. The structure is a sparse, tree-like skeleton with a few short cycles — consistent with  $80 - 69 + 1 = 12$  independent cycles — with two hubs of degree 7 and 6 near the centre and a long chain of degree-2 nodes reaching to the upper right (the component has diameter 17).

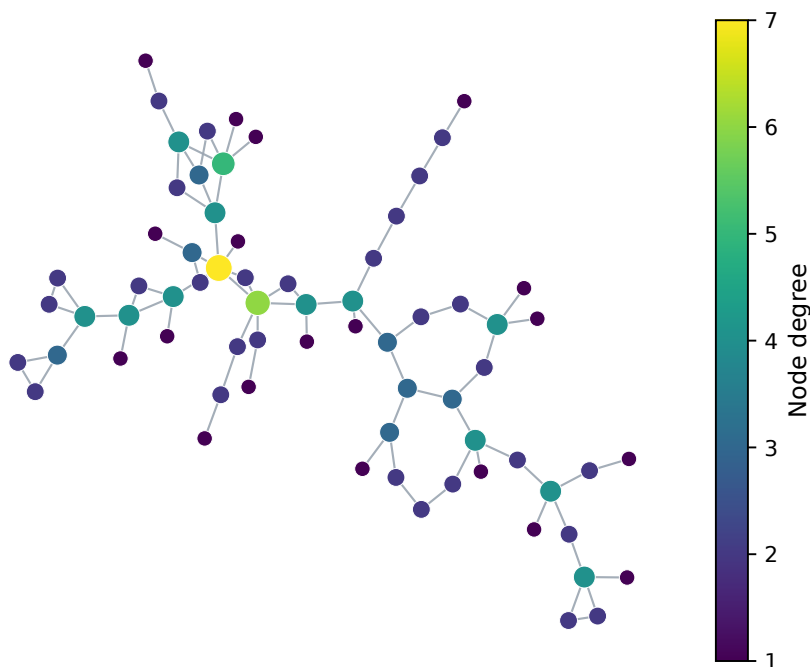


Figure 1: Q4(b): the second largest connected component (69 nodes, 80 edges), drawn with a Kamada–Kawai layout. Node size and colour both encode degree. Node labels are omitted for legibility.

#### Q4(c): Degree distribution of the whole graph (10 pts)

Figure 2 plots the degree distribution of the *entire* graph (all 26,728 nodes, not just the giant component). The left panel gives the raw node count  $N(k)$  on linear axes; the right panel gives the fraction of nodes  $P(k) = N(k)/n$  on log–log axes, which is the informative view for a sparse graph with a heavy tail.

**Answer.** The degree ranges from a **minimum of 1** to a **maximum of 59**, with **mean degree** 1.974 ( $= 2m/n = 2 \times 26,377/26,728$ ). No node is isolated, since every node enters the graph through an edge.

The distribution is strongly right-skewed. Degree 1 alone accounts for 13,942 nodes (52.2%) and degree 2 for a further 7,423 (27.8%), so four fifths of the graph has degree at most 2; the

| Degree statistic | Value |
|------------------|-------|
| Minimum degree   | 1     |
| Maximum degree   | 59    |
| Mean degree      | 1.974 |
| Median degree    | 1     |

Table 3: Q4(c): degree statistics over all 26,728 nodes.

median degree is 1. Only 37 distinct degree values occur at all, and the tail is thin and ragged — the largest degrees (36, 38, 40, 51, 57, 59) are each realised by a single node, the maximum being node 22777 with degree 59. On log-log axes the distribution falls off approximately linearly over the bulk of its range before scattering in the tail, which is the signature of a heavy-tailed, roughly power-law-like degree distribution rather than a Poisson one. This is also why the linear panel alone is misleading: on linear axes the whole distribution collapses into a single spike at  $k \leq 2$  and the tail out to  $k = 59$  is invisible.

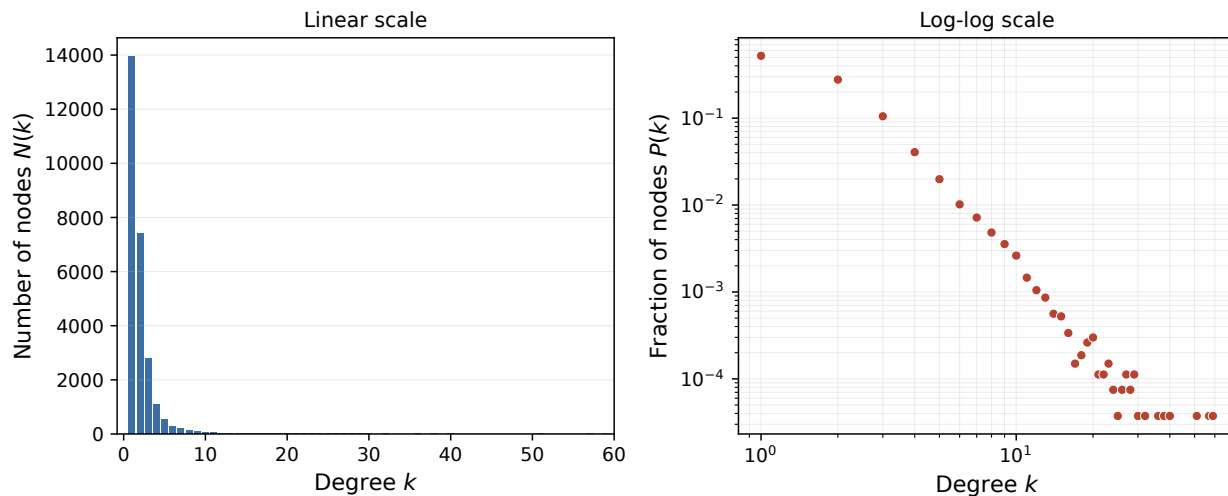


Figure 2: Q4(c): degree distribution of the whole graph. Left: node count  $N(k)$  against degree  $k$  on linear axes. Right: the same distribution as a fraction of nodes  $P(k)$  on log-log axes, which reveals the heavy tail out to  $k = 59$ .